Dennis Goyal

Software/Cloud Engineer

Software Developer and Data Engineer with 5 years of professional working experience. Specializing in Full-stack application development, DevOps, cloud-native technologies, with a strong research background in AI, cybersecurity, and compliance. Experienced in designing and implementing robust, fault-tolerant systems, and building intuitive, user-friendly interfaces that enhance user experience. Proven ability to lead and collaborate with cross-functional, and global, teams to deliver high-performance, scalable solutions.

Recognized for excellent communication, interpersonal skills, adaptability, and problem-solving abilities. Skilled in both high- and low-level programming paradigms, passionate about continuous learning and fostering productive teamwork. Accustomed to working with cross-cultural teams to achieve goals.

Professional Work Experience

April 2023 – August 2024

Cloud Software Engineer

Kempower Oyj, Finland

- Developed software across all layers of the EV charger management application - ChargEye with Angular frontend, serverless backend, AWS SQS, and DynamoDB.

- Planned and implemented new features in collaboration with the development team, product management, and stakeholders, using agile software development and DevOps practices.

- Worked with microservices architecture at scale with **AWS** cloud and **Serverless.js**, providing 1M+ charging sessions every month to end customers.

October 2023 – September 2024

Research Assistant

Scientific Advisory Board for Defence (MATINE)

- Adversarial Attacks and Defence strategies in ML based sensor fusion systems

- Explored and mitigated vulnerabilities in multi-modal fusion systems used in security and surveillance.

- Assessed system defences against 6 adversarial attacks like FGSM and AdvGAN, enhancing system robustness and cybersecurity of Machine Learning models through strategic defences and comprehensive training.

- Analysed multi-modality datasets and created LSTM and CNN models to validate defence effectiveness with **TensorFlow Keras** and Scikit-learn.

July 2020 – July 2022

Data Engineer

Elait Australia, Melbourne

- Data Governance, implementing robust frameworks to ensure data integrity, GDPR compliance and scalable solutions using **Azure** Containers.

- DevOps engineering with a focus on Continuous Integration and Continuous Deployment (CICD) methodologies using Azure, Terraform and Jenkins.

- Enterprise database design, Microsoft Business Intelligence tools to optimize data storage, retrieval, and analysis.

- Statistical analysis and data reconfiguration using **Databricks** (Apache Spark)

Contact

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Certified:

Azure Developer Associate / Azure Data Engineer

Technical Skills

Frameworks

Android and React Native

Microsoft.NET, Node.js, Django, Angular and React.js

Programming Languages

C/C++, Java, JavaScript, Python, SQL, HTML/CSS

Environments and IDEs

Windows and Linux systems, Docker containerisation, Kubernetes, ECS

Automation and DevOps

Terraform, Jenkins, GitLab CI/CD, Azure DevOps, Serverless.js, AWS Lambdas

Internet of Things

Apache Kafka, Edge computing, Redis and SQS

Real-time data processing and distributed systems in event-driven architecture

Cybersecurity

Network security protocols, Encryption standards, Firewalls, Penetration testing using OWASP ZAP and Burp Suite

Design and prototyping Tools

Figma, Adobe XD, Photoshop

Education

August 2022 – September 2024

Computer Science and Engineering

University of Oulu, Finland

- Obtained **Master's** degree in Computer Engineering with a focus on embedded systems, IoT, and signal processing.

- Developed strong expertise in cloud software, data engineering, and an understanding of how computing and information systems work under the hood.

- Graduated with a comprehensive understanding of state-of-the-art Artificial Intelligence, cybersecurity, and applied computing.

February 2016 – January 2020

Bachelor of Science

The University of Melbourne, Australia

- Obtained **Bachelor's** degree with a dual major in Electrical Systems and Computing and Software Systems and with a focus on Software Modelling and Design, Electrical Systems Modelling, Signals and Systems, Artificial Intelligence and Models of Computation

- Developed expertise in software engineering, programming, algorithms and data structures

- Gained in-depth knowledge of mathematical modelling, electrical systems, and applied engineering.

- Specialized in telecommunications, automation, energy distribution, and digital systems design

Leading Projects

Full stack development using Node.js – React and Express.

Jim's Removals – A removals job booking and ecommerce site.

- GitLab CI/CD, Node.js, PostgreSQL, React (Next.js), AWS Lambda
- Used Firebase Authentication (GCP), Saleor E-commerce, Stripe payments

Data Profiling and Exploratory analysis

Silver Chain Group Ltd

- Data Analytics and Profiling for Silverchain's Data Warehouse in **Azure Synapse Analytics**
- Developed efficient SQL Scripts to discover relationships across databases, tables, and columns, using data mining, inference and meta-data analysis
- Created visualisations for the relationships using Neo4j graph database
- Worked with Python Pandas, T-SQL, shell scripts, and data pipelines

Embedded Systems Project

University of Oulu

- Designed and built a robot integrated with sensors such as LiDAR, accelerometer, and Bluetooth module for obstacle detection and navigation
- Created custom PCB expansion board for STM32 microcontroller using KiCad and developed software for sensor control systems

Additional Experience

Cybersecurity researcher

Oulu University Secure Programming Group (OUSPG)

- Developed Java based open source Fuzzer plugin for OWASP ZAP web application security scanner; https://github.com/zaproxy/zaproxy

- Developed a full-stack LMS Hackathon type application in Angular/Node.js Express for a University course on Cybersecurity

Technologies: C/C++, Java, Git VCS, Angular, Swagger REST API docs, Linux administration, Sequelize ORM

Mathematics Tutor

Cluey Learning, Australia

- Taught 10+ Year 10, Year 11, and Year 12 students

- Excellence in curriculum across VCE, HSC and WACE

- Got excellent performance review for more than 70% of my students

Skills

Critical analysis and problem-solving
●●●●● Excellent

Object oriented design and ORM
●●●●○ Very Good

Algorithmic Programming and Data Structures
●●●●○ Very Good

Intelligent design, AI, and Machine Learning
●●●●● Excellent

Database design and development
●●●●● Excellent